

Applications of a General Theory of Architecture

Case Studies, Classifier Applications, and Implications

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Abstract

This paper applies the general theory of architecture to diverse domains. We demonstrate the classifier (tool vs game/relation) across physical, informational, organizational, and conceptual cases, showing context-dependence. We present a cross-domain set of examples, the Ethiopia test case, and the DRC Conflict Trap as a formalized case study. We also outline general applications (design, evaluation, diagnosis, forecasting, education) and practical implications for engineering, statecraft, and the limits of computation. Stability and desirability are independent; peace must be constructed.

Contents

1	The Classifier in Practice	1
2	Cross-Domain Examples	1
2.1	Physical Architecture: Hospital	1
2.2	Software Architecture: Web Service	2
2.3	Information Architecture: Library	2
2.4	Organizational Architecture: Corporation	2
2.5	Knowledge Architecture: Scientific Paradigm	2
3	Case Study: Ethiopia (Algorithm Test)	3
3.1	Formalization	3
3.2	Classification	3
3.3	Principles Applied	3
3.4	Constraints and Equilibria	3
4	Case Study: The DRC Conflict Trap	3
4.1	Actors	3
4.2	Utilities (Structural Form)	3
4.3	Mineral Rent and Double Stability	4
4.4	Indifference of External Actors	4
4.5	Conflict Trap Definition	4
4.6	Peace Condition	4
4.7	Five Architectural Interventions (Minimal Package)	4
4.8	Turing Barrier (DRC)	5
5	General Applications	5
6	Conclusion	5

1 The Classifier in Practice

The theory distinguishes tools (trivial induced game, unique computable equilibrium, narrow mode) from games/relations (non-trivial, extended mode or barrier). Classification is contextual and dynamic.

Algorithm 1 CLASSIFY_OBJECT (summary)

Require: X, E

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1: Identify parts  $P$ ; collect agents  $I = \{p \in P : \text{has\_own\_purpose}(p, X)\}$ 
2: if  $\exists i \in I : \text{is\_Turing\_complete}(i)$  then return undecidable (barrier)
3: end if
4: if  $\exists i \in I : \text{has\_private\_information}(i) \vee \exists i, j : \text{has\_commitment\_problem}(i, j) \vee \exists i :$ 
    $\text{can\_best\_respond}(i)$  then
5:   Build  $\mathcal{G}(X, E)$ , compute  $\mathcal{E}$ 
6:   if  $|\mathcal{E}| == 1$  and computable then return tool (narrow)
7:   elsereturn relation (extended)
8:   end if
9: end if
10: return tool (narrow)
```

Object	As tool	As game/relation
Hammer	Carpenter's tool	Weapon in fight
Gun	Hunter's tool	Relation soldier-enemy
Constitution	Legal text	Political game
Contract	Text	Relation between parties
Platform	Code	Market/game among users/regulators
Language	Grammar	Social interaction
Market	Mechanism	Game among buyers/sellers/regulators
Marriage	Contract/institution	Relationship between spouses
Road	Infrastructure	Game among drivers/pedestrians
Hospital	Building	Game among patients/staff/admin

Table 1: Same object, different modes by context.

2 Cross-Domain Examples

2.1 Physical Architecture: Hospital

Function: heal patients. Principles: F (recovery), σ (throughput), Σ (safety/security), K (beds/capacity). Environmental elements span passive (gravity, codes, seismic), passive living (sleeping patients), active non-living (devices/HVAC/robots), active living (staff/patients/visitors/microbes). Co-evolution: hospital shapes neighborhood; neighborhood shapes hospital. Constraints emerge from interaction (e.g. negative pressure isolation, redundant power, surge capacity). Trade-offs: wider corridors aid speed but reduce capacity; security vs access; beds vs efficiency.

2.2 Software Architecture: Web Service

Function: connect users/deliver content. Principles (five): latency, aesthetics/coherence of form/throughput/cost; security/resilience; scale/elasticity. Active elements include users, bots, competitors, regulators. Co-evolution: platform shapes discourse; discourse shapes platform.

Constraints emerge (e.g. low-latency global delivery with end-to-end encryption). Trade-offs: encryption vs searchability; caching vs freshness; horizontal scaling vs coordination overhead.

2.3 Information Architecture: Library

Function: preserve and provide access to knowledge. Principles (five): search/retrieval, efficiency, preservation/security, scale, aesthetics/legibility efficiency, preservation/security, scale of collections. Active elements include scholars, publishers, automated systems. Digitization improves access/scale but raises preservation/security challenges; physical slow but stable vs digital fast but fragile.

2.4 Organizational Architecture: Corporation

Function: create/deliver value. Principles (five): function (value creation), speed/efficiency, safety/security/risk, scale/capacity/load, aesthetics/organizational coherence. Active elements: employees, competitors, customers, regulators, automated systems. Trade-offs: centralization vs autonomy; flat vs hierarchical for scale. Organizational chart differs from actual functioning.

2.5 Knowledge Architecture: Scientific Paradigm

Function: explain and predict. Principles (five): explanatory/predictive power (function), productivity/efficiency, robustness/correctability (safety), breadth/depth (scale), aesthetic/conceptual coherence (beauty/symmetry) power (function), research productivity/efficiency, error correction/robustness, breadth/depth (scale). Co-evolution: paradigm shapes questions asked; answers reshape paradigm (Kuhn). Constraints include falsifiability, reproducibility, consilience, mathematical coherence.

3 Case Study: Ethiopia (Algorithm Test)

3.1 Formalization

Goal: “Ethiopia unified, peaceful, and prosperous.” Utility $u_{\text{total}} = w_1 u_{\text{unity}} + w_2 u_{\text{peace}} + w_3 u_{\text{prosperity}}$. Terms are contested; analysis is Pareto exploration.

3.2 Classification

Parts (ethnic groups, political parties, regional governments, religious institutions, foreign powers) have purposes of their own, private information, commitment problems, and strategic response capacity. Classification: **relation** (game); mode: **extended**.

3.3 Principles Applied

Principle	Measurement/concerns
Function	Territorial integrity, inter-group trust, service delivery
Speed/Efficiency	Policy implementation time, corruption friction
Safety/Security	Risk of state collapse, civil war, famine
Scale/Capacity/Load	120M people, 80+ groups, 1.1M km ²
Aesthetics/Beauty/Symmetry	Legitimacy, symbolic coherence, unity in diversity

3.4 Constraints and Equilibria

Constraints inferred include: imposed unity generates resistance; multi-ethnic stability needs common identity or protective federalism; without credible commitment groups arm (conflict equilibrium); growth without redistribution intensifies grievance; architecture viable in hostile regional environment; commitment/time inconsistency, federalism paradox, elite capture, diaspora/digital amplification.

Induced game has multiple equilibria including conflict traps. Security dilemma, commitment problem, and external interests generate stable undesirable equilibria. Peace is not default; it must be constructed. Utility is contested, parts contested, environment adversarial, feasible set not fixed, probes costly (human). Value is diagnostic, not prescriptive.

4 Case Study: The DRC Conflict Trap

4.1 Actors

Principal actors: Kinshasa K (DRC state), M23/AFC M (rebel coalition), Rwanda R , external actors X (US, China, Gulf states, UN/AU arenas). Architecture determines available strategies.

4.2 Utilities (Structural Form)

$$u_K \approx \alpha_K K + \beta_K K + \gamma_K K - \delta_K K, \quad (1)$$

$$u_M \approx \alpha_M M + \beta_M M + \gamma_M \text{Leverage}_M \quad (\text{instrumental}). \quad (2)$$

Both dominated by territorial control, mineral extraction, external patronage, subject to military cost.

Proposition 4.1 (Identity of practical goals). *Under current architecture, $u_K \approx u_M$ structurally (same form, different beneficiary). Conflict is competition for control of the architecture, not contest over what it should be.*

4.3 Mineral Rent and Double Stability

Resource rent $\Pi = V(\mathcal{M}) \cdot \rho \cdot (1 - \lambda)$. Conflict stable if $\Pi(i) > \Pi(i)$ for pivotal i ; in DRC holds for both $\rightarrow$ doubly stable.

Proposition 4.2 (Double stability). *Neither side wants to destroy architecture (both want to inherit); neither can credibly commit to power-sharing (prize indivisible). Result: war of position, not transformation.*

4.4 Indifference of External Actors

Proposition 4.3 (Indifference). *External actors have no interest in peace/war as such; $u_X() \geq u_X()$ by parameters. Peace requires payoff redesign so utility-maximizing behavior produces peace as byproduct (not persuasion).*

Examples: US pro-access (critical minerals/supply chain), China pro-supply, Rwanda pro-Rwanda calculus, Gulf states pro-influence, UN/AU constrained by members.

4.5 Conflict Trap Definition

Conflict trap (A, s^*) with $s^* \in \mathcal{E}$, undesirable outcome $\varphi_A(s^*, E) \notin G$, Harm > 0 , robust. Four classes map to stability-desirability.

4.6 Peace Condition

Peace is stable equilibrium iff:

$$\begin{cases} \Pi(K) > \Pi(M), \\ \Pi(M) > \Pi(K), \\ u_R() > u_R(), \\ u_X() > u_X() \forall \text{pivotal } X, \\ \text{credible commitment devices exist.} \end{cases} \quad (3)$$

Negotiation within existing architecture fails because architecture is prize, not stake.

4.7 Five Architectural Interventions (Minimal Package)

Pillar	Intervention	Actors	Mechanism
Divisible rent	Federalism + revenue sharing	K, M	Make prize shareable
Architecture change	Green Corridor / productive economy	All	Create economic alternative to extraction
Credible commitment	Verification mechanisms	All	Reduce commitment problem
External pay-off design	Conditional access + sanctions	X	Make peace more profitable than conflict
Path dependence	National dialogue / address inheritance	All	Address constraint inheritance

Table 2: Minimal peace package.

4.8 Turing Barrier (DRC)

Actors are Turing-complete $\rightarrow$ no algorithm can decide whether peace equilibrium exists or which equilibrium selected. Theory maps structure/conditions/interventions but cannot guarantee outcome.

5 General Applications

- **Design:** heuristics for any architecture (purpose, parts/constraints, environment passive/active/living/non-living, measure principles, design for co-evolution).
- **Evaluation:** audit existing architectures against $(F, \sigma, \Sigma, K, -\rho)$ and emergent constraints; identify what must be respected vs shaped.
- **Diagnosis:** locate failure (function, speed, safety, scale, ignored parts/constraints, misclassified active/passive, living/non-living confusion).
- **Forecasting:** anticipate problems as load/threats/purpose/parts/constraints evolve; design for adaptation/learning.
- **Education:** substrate-independent framework bridging engineering, organizational design, knowledge work, policy/statecraft.

6 Conclusion

Applying the theory across domains shows: classification (tool vs game/relation) is contextual and dynamic; same object can occupy different modes. Ethiopia illustrates diagnostic value in contested multi-agent settings (extended mode). The DRC case formalizes a mineral rent conflict trap where incumbents and rivals share structurally similar utilities (competition for control, not contest over architecture) and external actors are indifferent to peace/war as such—so peace requires payoff redesign, not persuasion. Across cases, stability and desirability are independent; peace must be constructed via architecture that induces desirable stable equilibrium. Conflict is the generic case; peace is the special case requiring explanation.